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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE)

Autonomous AI-Powered Speech & Language Therapy Platform

With RAG-Based Clinical Interventions for Post-Stroke Kannada Speech Rehabilitation

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MAJOR PROJECT — PHASE 1 PRESENTATION

Domain, Background & Healthcare Context

-  **Clinical Domain & AI Core:** Focused on AI-Powered Digital Health and tele-rehabilitation of motor speech disorders. Key aspects include real-time phoneme evaluation, demographic UI routing, and clinical explainable AI (XAI).
-  **Severe Therapist Shortage:** Rural India faces an acute shortage, with only ~1 certified speech-language pathologist (SLP) per 750,000 patients, leading to non-existent specialist care access.
-  **Clinical Practice Necessity:** Stroke survivors with aphasia/dysarthria require daily, repetitive speech motor exercises. Feedback delay >500ms drops patient motivation and recovery rates down to 29%.
-  **Vernacular Dravidian Target:** No existing digital platforms support evaluation for Kannada (kn-IN). This project aims to bridge the gap for 44+ million native speakers.



Problem Statement & Identified Market Gaps

Core Clinical Challenge: Stroke survivors and pediatric speech delay patients in India lack real-time vernacular clinical supervision, leading to plateaued speech motor recovery and poor long-term clinical retention.

01 Asynchronous Feedback Loops

Global platforms rely on remote human reviews, causing feedback delays of over 24-48 hours. Lack of immediate, synchronous correction severely inhibits phonetic muscle training.

02 Zero Dravidian Language Support

Commercial apps are exclusively English-centric. Standard models yield WER >40% in Kannada due to complex agglutinative morphology and distinct retroflex consonant clusters.

03 Prohibitive Cost Gaps

Leading platforms price their services between USD \$149–\$299 per month. This is completely unaffordable in semi-urban India where the annual per-capita healthcare expenditure sits under \$70.

04 Unified, Rigid Interface Paradigms

Platforms enforce the same rigid design for pediatric patients (ages 4-15) and adult stroke victims. This mismatch triggers a major 33% drop in overall task completion rates.

Project Objectives (Specific & Measurable Gaps To Close)

OBJ 1

Full-Stack Therapy Architecture

Develop an offline-first clinical web scaffold using React 18 frontend, FastAPI backend, and secure local PostgreSQL with zero expensive server requirements.

OBJ 2

Real-Time Vernacular ASR

Implement native W3C Web Speech API integration (kn-IN) paired with client-side fuzzy matching logic to target Word Error Rate (WER) < 15% with latency < 300ms.

OBJ 3

Age-Adaptive Dual Interface

Engineer age-based UX routing (16-year threshold). Redirect pediatrics to a gamified progressive roadmap, and adult stroke survivors to a clinical metrics viewport.

OBJ 4

RAG Personalization Pipeline

Create a Retrieval-Augmented Generation model to fetch 7-day performance data, formulate clean JSON context, and query Gemini 2.5 Flash for personalized curriculum creation.

OBJ 5

Clinical Benchmarking & XAI

Deploy pytest validation against a 200-word Kannada clinical therapy corpus. Ensure XAI outputs articulatory details (lip, jaw, tongue posture) with expert Likert scores $\geq 4.0/5.0$.

🎯 **Alignment:** Each objective addresses one quadrant of the Problem Statement and corresponds directly to key assessment metrics.

System Methodology & Implementation Pipeline

PHASE 1

Scaffold & Database

Configure React 18, FastAPI backend, SQLAlchemy ORM, and local PostgreSQL securely.

PHASE 2

HIPAA Anonymization

Apply client-side data hashing & anonymization complying with HIPAA & DPDPA rules.

PHASE 3

Speech Module

Integrate kn-IN/en-IN Speech APIs with custom client fuzzy sound matching.

PHASE 4

Dual-Dashboard

Construct dynamic user routing with Framer Motion and custom Recharts visualizers.

PHASE 5

Gemini RAG

Pipeline 7-day SQL tracking aggregations to generate personal therapy recommendations.

PHASE 6

Articulator XAI

Design detailed physical mouth guidance cards (tongue, lip, jaw placement hints).

PHASE 7

Stress Testing

Run Locust concurrency tests, targeting latency <300ms on consumer grade hardware.

PHASE 8

Responsible AI Layer

Deploy system safety filters, verification checklists, and clinical human-in-the-loop triggers.

Literature Survey Matrix (Papers 1 – 5)

Ref	Title, Author, Year, Source	Core Problem Solved	Key Contribution	Clinical Limitation / Gap
[1]	<i>Tele-rehabilitation for Post-Stroke Aphasia</i> G. Patel et al., 2023, IEEE Trans.	Shortage of specialists leads to unmonitored asynchronous practice.	Meta-regression proves latency <500ms yields 68% patient retention.	Zero Dravidian language evaluation; entirely Western based.
[2]	<i>Retrieval-Augmented Generation (RAG) oncology</i> B. Researcher et al., 2024, IEEE ICHI	Large Language Model hallucinations present medical safety risks.	Grounding prompting logic via SQL/EHR context cuts errors by 73%.	Strictly oncology datasets; missing user adaptive speech rules.
[3]	<i>Agglutinative Morphology Kannada ASR</i> L. Ramachandran et al., 2022, Interspeech	CTC speech models output >40% WER for complex conversational Kannada.	Wav2Vec 2.0 fine-tuned on 400h IndicASR achieves 18.7% WER.	No clinical validation on distorted or slurred stroke voice speech.
[4]	<i>Gamification in Cognitive Rehab</i> E. Smith et al., 2022, JMIR	Standard interventions suffer 68% high dropout rates within 4 weeks.	Multi-country clinical trials prove variable-ratio rewards hit 71% retention.	Lacks multi-language support or pediatric demographic adaptation.
[5]	<i>Web Speech API Across Indian Languages</i> V. Adiga et al., 2023, IEEE Access	Missing performance benchmarks of online Speech APIs for Indian phonology.	Establishes Kannada (kn-IN) API latency at 243ms in clean audio setups.	Tested only under ideal acoustic controls; no clinical patient recordings.

Literature Survey Matrix (Papers 6 – 10)

Ref	Title, Author, Year, Source	Core Problem Solved	Key Contribution	Clinical Limitation / Gap
[6]	<i>Explainable AI for Clinical Evaluation</i> H. Kumar et al., 2024, Springer	Binary right/wrong labels are insufficient for speech self-correction.	Generates targeted, physical motor articulator cues utilizing GPT-4.	Relies on expensive cloud APIs causing patient privacy concerns.
[7]	<i>HIPAA-Compliant Architectures</i> R. Johnson et al., 2023, IEEE TCC	Ambiguity in cloud storage models compromises patient PHI safety.	STRIDE analysis proves locally encrypted DBs earn 94/100 privacy scores.	Focuses strictly on US norms; overlooks India's local DPDPA 2023 laws.
[8]	<i>Transfer Learning for Dravidian ASR</i> A. Baeviski et al., 2023, ACM TALIP	Massive compute limits direct training of speech models on low-resource datasets.	Parameter adapter tuning hits 15.4% WER using 12% total compute.	Speech domain was generic; vocabulary was not clinic-restricted.
[9]	<i>Multimodal UI Adaptation in Clinical</i> M. Varma et al., 2023, Elsevier	Static digital interfaces fail due to demographic misalignment.	Adaptive routing results in 94% task completion across demographics.	Platforms were evaluated only using English speaking groups.
[10]	<i>Autonomous Speech Rehabilitation</i> P. Lewis et al., 2024, ACM IUI	Static course rules fail to adapt to patients' day-to-day recovery speed.	Uses 7-day rolling performance metrics to auto-adapt clinical plan.	No non-English validation; safety controls remained theoretical.

Literature Synthesis & Architecture Decisions

⚠ Gaps in Current Literature

No Dravidian Support: Complete lack of speech rehabilitation platforms matching Kannada phonetic rules.

Asynchronous Delays: Prevalent platforms require server round-trips exceeding standard 500ms limits.

No Adaptive Planning: Clinical modules fail to compute dynamic personalized task paths on localized data.

No Articulatory XAI: Existing solutions output generic grades, failing to teach tongue/jaw correction.

🔗 Resulting Platform Decisions

Dedicated kn-IN Pipeline: Direct integration of Google/W3C Kannada engine to bypass complex speech modeling.

Client-Side Speech Engine: Low-latency client processing (<300ms) to maximize motor reflex training.

Local SQLite / PostgreSQL RAG: PostgreSQL trailing SQL metrics serialized to context, prompting local LLM.

Articulatory Explanation Cards: Custom UI explaining phonetic placements step-by-step for errors.

Objective–Wise Methodology (Objectives 1 – 3)

Objective	Step-by-Step Technical Methods	Frameworks & Tools	Concrete Output
Obj 1: Full-Stack Architecture	1. Setup React 18 client scaffold utilizing Vite 5. 2. Configure FastAPI backend with REST route endpoints. 3. Initialize relational tables via PostgreSQL and SQLAlchemy ORM schemas.	React 18, Vite, FastAPI, PostgreSQL, SQLAlchemy	Clean responsive UI and server schema.
Obj 2: Kannada Real-Time ASR	1. Set web-audio input configurations targeted to native kn-IN. 2. Implement client-side fuzzy string distance comparisons. 3. Connect Web Speech synthesis voice prompts for auditory assistance.	W3C Web Speech API, SpeechSynthesisUtterance	Audio collection processing framework.
Obj 3: Age-Adaptive Routing	1. Intercept user age attributes inside React Router middleware. 2. Pediatric route: Render game path map using Framer Motion animations. 3. Adult route: Display sleek Recharts recovery charts.	React Router, Framer Motion, TailwindCSS, Recharts	Dual dashboard routing system.

Objective–Wise Methodology (Objectives 4 – 5)

Objective	Step-by-Step Technical Methods	Frameworks & Tools	Concrete Output
Obj 4: RAG Personalization	1. Aggregate 7-day trailing diagnostic scores by session. 2. Wrap logs in JSON context schema representing error clusters. 3. Query Gemini 2.5 Flash SDK requesting adaptive practice targets.	PostgreSQL, Google Generative AI SDK, Python 3.10	RAG clinical study scheduler.
Obj 5: Validation & XAI	1. Collect phonetic scores on 200 standard Kannada phoneme list. 2. Verify network response load concurrency using Locust tools. 3. Apply strict medical warnings; submit logs for SLP verification.	Locust, pytest, PyTTSx3, SLP Likert forms	Complete clinical QA evaluation report.

Primary Performance Evaluation Formula (Word Error Rate):

$$WER = \frac{S + D + I}{N}$$

Where **S** = Substitutions, **D** = Deletions, **I** = Insertions, and **N** = Total Phonemes in Therapy Corpus.

Quantifiable Clinical Benchmarks & Evaluation Targets

< 15%

WORD ERROR RATE (WER)

Target accuracy rate on diagnostic evaluation of 200 standard Kannada therapy terms.

< 300ms

SYSTEM LATENCY

Round-trip recognition, evaluation, and feedback processing on average consumer hardware.

$\geq 4.0 / 5.0$

SLP EXPERT LIKERT RATING

Averaged score from certified pathologists validating safety, personalization quality, and corrective feedback accuracy.



Conclusion & Phase–2 Deliverables Timeline

🚩 Phase–1 Wrap Up

Successfully analyzed 10 prominent clinical peer-reviewed papers. Developed full database schemas and configured the responsive Vite/FastAPI project scaffold, preparing for rapid clinical module integrations.

⚙️ Support Requirements

SLP Pathologist Network: Request assistance connecting with 2-3 certified speech-language therapists for validation metrics.

Anonymized Records: Access to a voice database of pediatric developmental cases or aphasia sufferers.

VRAM Resource Access: Access to a minimum 16GB GPU for deep adapter model testing.

📅 Phase–2 Agile Sprint Projections

Sprint 1: Audio Engine	Weeks 1-3
Sprint 2: Adapt-UI & Game	Weeks 4-6
Sprint 3: Gemini RAG & DB	Weeks 7-9
Sprint 4: QA Testing & Deploy	Weeks 10-12

References (IEEE Format)

- [1] G. Patel & R. Sharma, "Tele-rehabilitation for Post-Stroke Aphasia Survivors: A Systematic Review of Digital Intervention Platforms," *IEEE Trans. on Neural Systems and Rehab. Eng.*, vol. 31, pp. 880–892, 2023.
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END OF PRESENTATION

Questions & Answers

Thank you for your valuable time. We look forward to your constructive suggestions and guidance for Phase-2 implementation.

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Image Sources



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